TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester
Semester: V
Programme: B. Tech Information Technology
Course: Advanced Programming with Java

Time: 3 Hours **Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33501	Course Outcome
CO1	To classify core Java fundamentals.
CO2	To analyze arrays, packages, and interfaces to develop modular and reusable Java applications.
CO3	To develop and manage multithreaded applications.
CO4	To utilize inheritance and polymorphism principles to create scalable Java applications.
CO5	To demonstrate knowledge of JDBC for database connectivity.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define JVM.	CO1	2	2
b.	What is array?	CO2	2	2
c.	Define thread.	CO	2	2
d.	What is polymorphism?	CO	3	2
e.	Define JDBC.	CO	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain Java data types and operators.	CO1	2	5
b.	Explain control structures.	CO1	4	5
c.	Explain collections framework.	C01	3	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain arrays with program.	CO2	2	5
b.	Explain interfaces with example.	CO2	3	5
c.	Explain exception handling with program.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain thread creation methods.	CO3	2	5
b.	Explain synchronization with example.	CO3	4	5
c.	Explain thread life cycle.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain inheritance with example.	CO4	4	5
b.	Explain method overriding.	CO4	3	5
c.	Explain types of inheritance.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain JDBC architecture.	CO5	2	5
b.	Write steps to connect Java with database.	CO5	3	5
c.	Explain JDBC driver types.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create